

Mahesh Basavaraj

AI RESEARCHER · EDUCATOR

CONTACT

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AT A GLANCE

11 peer-reviewed papers

4 patents & applications

4 journal articles

7 conference papers

RESEARCH AREAS

- Agentic AI & LLM systems
- Spatio-temporal & attention models
- EEG & brain–computer interfaces
- Cognitive IoT & cross-layer optimisation
- Responsible AI & fairness
- Anomaly detection

Research Publications & Patents

PREPARED FOR THE ASSISTANT PROFESSOR (COMPUTER SCIENCE) ROLE — UNSW BENGALURU CAMPUS

RESEARCH PROFILE

My research asks how intelligent systems can be made reliable and useful outside the laboratory: how agentic systems built on large language models coordinate to solve multi-step problems, how deep models recover structure from noisy spatio-temporal signals, and how constrained networked devices share resources fairly. The work is deliberately applied. It has produced three utility patents and a published application, and it is shaped by eight years of building production software before I moved into academia. Current threads are the evaluation and reliability of LLM-based agents, attention architectures for EEG and motor-imagery decoding, cross-layer optimisation in heterogeneous IoT, and fairness in deployed machine-learning models.

SELECTED CONTRIBUTIONS

- Multi-scale spatio-temporal attention for motor-imagery decoding — a network combining frequency–spatial parallel attention with multi-scale temporal modelling (IEEE ICIICS 2026).
- Lightweight multi-agent Q-learning for wireless IoT — cross-layer optimisation that balances energy across heterogeneous devices without central coordination (IEEE ICMNWC 2025).
- Agentic LLM systems as a route to general intelligence — a collaborative multi-agent framework analysed against SayCan, ChatDev and Devin (INDJCST 2025).
- Fairness in deployed models — a patented framework for bias mitigation and fairness verification in machine-learning pipelines (2026).

JOURNAL ARTICLES

Medical Science Educator **Comment on “Incorporating Artificial Intelligence in Qualitative Research: Exploring the Role of ChatGPT in Thematic Analysis”**
Vol. 36, Issue 2, pp. 1063–1064 · 2026
DOI 10.1007/s40670-025-02618-y
with S. Banerjee, I. H. Madar

J. Neurological Sciences **Comment on “Improving care for amyotrophic lateral sclerosis with artificial intelligence and affective computing”**
Vol. 486 · 2026
with S. U. Adhyapak, P. Mishra

INDJCST **Agentic Approach in the Quest for AGI**
Indian Journal of Computer Science and Technology, pp. 279–289 · 2025
DOI 10.59256/indjcst.20250402043

IRJET **A Review of Deep Learning Techniques in Detection of Anomaly in Credit Card Transactions**
Vol. 9, Issue 9 · 2022

PATENTS

01 **A Bulk Silkworm Cocoon Inspection and Assisted Sorting Machine Using Shallow-Layer Conveyor Spreading**
Utility patent · 2026

02 **Adaptive Quality of Service Management System for Heterogeneous Internet of Things Networks**
Utility patent · 2026

03 **Ethical and Responsible AI: Bias Mitigation and Fairness in Machine Learning Models**
Utility patent · 2026

04 Cloud-based Social Virtual Reality IoT Network with Device Synchronization Between Real and Virtual Worlds
Published application 202441098998 · filed 14 Dec 2024, published 20 Dec 2024

DOCTORAL RESEARCH

Ph.D. in Computer Science & Engineering, Visvesvaraya Technological University (enrolled 2023), in applied deep learning and intelligent systems. Supervised through DSATM, Bengaluru.

RESEARCH TRAINING

Deep Learning — Elite
NPTEL · 2025

Blockchain Technology
NPTEL · 2025

Data Analytics with Python
NPTEL · 2024

Machine Learning & NLP
Infosys Springboard FDP · 2025

Cyber Security
AICTE · VTU FDP · 2023

REPRODUCIBILITY

Research code and teaching systems are public at **github.com/maheshbasavaraj**

CONFERENCE PAPERS

ICIICS 2026 **Multi-Scale Spatio-Temporal Network with Frequency–Spatial Parallel Attention for Motor Imagery Classification**
3rd Intl. Conf. on Integrated Intelligence and Communication Systems, pp. 1–7 · May 2026
DOI 10.1109/ICIICS67880.2026.11483407
with R. A. M. Sait, B. Y. Vineetha, G. Priyadharshini, N. Sitanshu

ICIICS 2026 **Predictive Mobility-Aware Cognitive Medium Access Control for Underwater Vehicle Communication**
3rd Intl. Conf. on Integrated Intelligence and Communication Systems · 2026
with P. Maderamitla, J. Ulaganathan, G. Bindu, K. Anguraj

ICBDML 2026 **Harnessing Large Language Models for Next-Generation Artificial Intelligence Applications**
2nd Intl. Conf. on Big Data and Machine Learning · Feb 2026
with N. Garg, S. Yangalasetty, T. Vangalapat, M. Kumari, M. Gupta

IEEE ICMNWC 2025 **Lightweight Multi-Agent Q-learning Cross-Layer Optimization in Wireless IoT Networks for Balanced Energy Utilization**
IEEE Intl. Conf. on Modelling, Networks and Wireless Communication · Dec 2025

ICEASD 2024 **Using AI Agents Built on LLMs for Building AGI**
Intl. Conf. on Engineering Advancements and Sustainable Development, JETIR Vol. 11, Issue 5 · May 2024

ICAMSnT 2023 **The Alignment Problem in AI**
Intl. Conf. on Advances in Modern Science and Technology · Apr 2023

ICAPCTA 2022 **Evaluating OpenAI Codex for Real-World Coding**
Intl. Conf. on Advances in Parallel Computing Technologies & Applications · Jan 2022

RESEARCH SUPERVISION

Final-year capstone and project supervision since 2021 (DSATM, City Engineering College, T. John Institute of Technology), across applied machine learning, data engineering and intelligent systems. Teams are expected to produce a working artefact, a written report and a defensible evaluation — several capstones have grown out of the open-source teaching systems listed overleaf.

RESEARCH COMMUNICATION & OUTREACH

I write and speak about AI for a general technical audience, which keeps my research legible outside the conference circuit.

- **The Ghost in the Feed: Inside the First AI Civilization**
- **Thermodynamic Computing: Extropic's Chip and the Path to Efficient AI**
- **The Patent Paradox: AI and the Future of Intellectual Property**
- **Indian Academia's Obsolete Secret**
- **Scientific Papers are DOA**

Video explainers on the **Echoes of AI** YouTube channel.

COLLABORATION INTERESTS

Open to joint work on agent evaluation and LLM reliability, spatio-temporal modelling for biomedical signals, QoS in large-scale IoT deployments, and applied fairness auditing — particularly with industry partners who can supply real data and deployment constraints.